The triangle inequality

1

a

i

- $7 + 12 > 17$
- $7 + 17 > 12$
- $17 + 12 > 7$

ii Yes

b

i

- $3.2 + 7.4 > 9.6$
- $3.2 + 9.6 > 7.4$
- $9.6 + 7.4 > 3.2$

ii Yes

c

i

- $9 + 4 > 5$
- $9 + 5 > 4$
- $5 + 4 = 9$

ii Yes

d

i

- $3.1 + 7.7 < 11.8$
- $3.1 + 11.8 > 7.7$
- $11.8 + 7.7 > 3.1$

ii No

2

a $8 + 8 > 8$

b Yes

3

a Yes

b No

c No

4 The angle sum exceeds 180 degrees.

5

a $9 < a < 43$

b $7.4 < a < 41$

c $7 < a < 35$

d $17.6 < a < 75.8$

e $12 < a < 44$

f $0.8 < a < 4$

g $10 < a < 44$

h $7 < a < 43$

Additional questions

6 The sum of the lengths of any two sides of a triangle is greater than the length of the third side.

$AB + BC > AC$ and $AB + AC > BC$ and $AC + BC > AB$

7

a Valid

b Not valid, $4 + 7 \not> 12$

c Not valid, $6 + 7 \not> 13$

8

a

i $17+26>a$, so $43>a$
$17+a>26$, so $a>9$
$26+a>17$, so $a>-9$, which is true if $a>9$

ii $9<a<43$

b

i $15+18>a$, so $33>a$
$15+a>18$, so $a>3$
$18+a>15$, so $a>-3$, which is true if $a>3$

ii $3>a>33$

9 $-\frac{3}{5}<x<11$

10 He is correct that a valid triangle does not exist. The two doors just touching and forming a triangle would be a side length of $2 \cdot 36=72$ inches. The other two sides are 16 inches and $4.5 \cdot 12=54$ inches. However, $16+54 \ngtr 72$, so this is not a valid triangle.
His error is that just because they do not form a triangle, we cannot conclude that the doors do not touch. We can give a few different reasons for this:

- Although, the range of values that forms a triangle is $38<x<70$, this is a right triangle, so there is actually only one valid triangle with hypotenuse length of 56.32- which can be calculated with the Pythagorean theorem
- The doors will not ever touch if the sum of the two doors is less than the hypotenuse, which is not the case
- Since it is over the hypotenuse, the two doors will touch and form a quadrilateral

11 This is not a valid conclusion. This incorrectly assumes that Jacksonville, Orlando and Miami are all collinear. This is not necessarily the case, so they would form a triangle with vertices J for Jacksonville, O for Orlando, and M for Miami. Using the triangle inequality so we know that:
$JO+OM>JM$ or that $OM>JM-JO$, so the distance from Orlando to Miami is greater than 204 miles.